Paul Witkowski | AI Engineering Student in M2 specialised in Computer Vision

paulwitkowski.com | linkedin.com/in/paul-stanislas-witkowski | GitHub: Pauwit

Languages: French (native), German (native), English (C1)

Open to a **6-month internship** from **February 2027** in Machine Learning / Computer Vision

SKILLS

AI & ML	PyTorch, Hugging Face, MLflow, LoRA, vLLM, scikit-learn, NumPy, Pandas
Computer Vision	U-Net/U-ResNet, YOLO11, segmentation, image registration, ONNX, FFTW
MLOps & Infra	Docker, Kubernetes (k3s), Azure ML & Custom Vision, ONNX, Git, GitLab
Programming	Python, C, C++, C#, Java, Scala, OCaml, TypeScript, CUDA

EDUCATION

EPITA, MSc Data Science & Artificial Intelligence (SCIA) — Le Kremlin-Bicêtre, France
Top 4% of my year in the integrated preparatory cycle (2022–2024) — 2022 – 2027 (graduating)

Boston University, Exchange Semester — Boston, USA
Computer Science & Engineering — Jan 2024 – May 2024

Lycée Franco-Allemand, French-German Baccalaureate (**AbiBac**), with honours — Buc, France | 2022

EXPERIENCE

AI Research Engineer (End-of-Studies Research Project, PFEE) — Paris, France
Cloud Temple — Sept 2026 – Jan 2027

- Building a **sovereign AI factory** end to end: secure **GPU inference infrastructure** plus training and evaluation pipelines, scaling from **LoRA** → **full SFT** → **continued pre-training**.
- Scaling to an **80B-parameter model** with multi-GPU sharding; a custom **Green AI orchestrator** schedules inference and training workloads to cut compute cost.

Deep Learning Research Intern, Medical Computer Vision — Paris, France | **GitHub**
Institut Langevin (CNRS) / Quinze-Vingts Hospital — Sept 2025 – Jan 2026

- **Model engineering in Python/PyTorch:** designed and trained CNNs (**U-Net/U-ResNet**, then **YOLO11**) for segmentation and classification of retinal biomarkers in **laser Doppler holography**, on **805 annotated clinical images** with strict subject-level splits: **Dice 0.939**, **sub-2% failure rate**.
- **Production deployment:** exported models to **ONNX** and integrated them into the lab's MATLAB software *EyeFlow*, running offline on-premise for **medical-data privacy**; models published on Hugging Face and **used daily in hospitals**.
- **Annotation tooling:** wrote a **GIMP 3 Python plugin** turning the save-mask / open-next annotation cycle into a single keystroke, used to label the training set.
- **First-author paper** (in preparation) on automated retinal calibration and laterality classification, with an **uncertainty budget** isolating algorithmic (5.16%) from biological (1.35%) error.

Teaching Assistant, C# & OCaml — Le Kremlin-Bicêtre, France
EPITA — Sept 2024 – June 2025

- Mentored **first-year students** in algorithmics and programming; designed labs, ran evaluations and code reviews.

PROJECTS & HACKATHONS

EpiTweet: Distributed Micro-Blogging Platform — EPITA, Team of 14 | private repo

- Owned the **user-service** in Java/Quarkus: the **Neo4j** social graph, MongoDB profiles and **Redis pub/sub** events. Every merge request had to pass our CI and a **full test suite with coverage checks** before rolling out to **Kubernetes (k3s)**.

Diffeomorphic Demons: Medical Image Registration in C++ — Institut Langevin internship | **GitHub**

- **C++ port** of the Diffeomorphic Demons algorithm, aligning two medical images through a **smooth, invertible deformation field**; computations moved to the **frequency domain with FFTW** to sharply reduce complexity, validated against the reference implementation.

StockAI: Intelligent Shelf Restocking – Ranked 1st of the Course — EPITA, Group of 4 | **GitHub**

- Gradient-boosting demand forecaster tracked in **MLflow**, registered in **Azure ML** and served from a **managed real-time endpoint**; **Azure Custom Vision** shelf detector runs on Azure or as an exported **edge container**; an **LLM agent** queries stock over **MCP** to decide restock vs. supplier order.

Neural Network from Scratch — Personal project | **GitHub**

- MNIST digit classifier in **C++ with no ML libraries**: hand-rolled matrix class, feedforward, **backpropagation** and **mini-batch SGD**, with checkpoint save/load and switchable sigmoid/LeakyReLU activations.

VOLUNTEERING

Volunteer & Workshop Instructor — Paris, France
Prologin – Student Association — Jan 2026 – present

- Leading coding workshops for school students; organised two national coding competitions (**1,000+ entrants**).